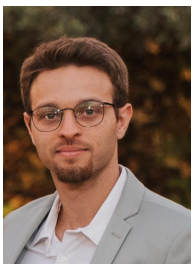
Guy Bar-Shalom

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https://barsguy.github.io/



Education

2022 – 2026 **PhD Computer Science.**
Technion - Israel Institute of Technology.
Advisors Prof. Ran El-Yaniv, and Prof. Haggai Maron.
Graph Neural Networks.

2020 – 2022 **MSc Computer Science.**
Advisor Prof. Ran El-Yaniv.
Technion - Israel Institute of Technology.
Uncertainty Estimation in Deep Learning Models.

2016 – 2020 **BSc Computer Science and Physics.**
Technion - Israel Institute of Technology.

Employment

2025 **Meta.**
Research Intern
Stating in May 2025

2023 **Google-Verily.**
Research Intern
Video analysis (published at WACV)

2018 – present **Teaching assistant (Technion).**
Algorithms 1.
Partial differential equations and Fourier series.
Introduction to Computer Science - C.

Selected Publications (* indicates equal contribution)

2025 Learning on LLM Output Signatures for Gray Box LLM Behavior Analysis
Guy Bar-Shalom*, Fabrizio Frasca*, Derek Lim, Yoav Gelberg, Yftah Ziser, Ran El-Yaniv, Gal Chechik, Haggai Maron
(Submitted to ICML)

Balancing Efficiency and Expressiveness: Subgraph GNNs with Walk-Based Centrality
Joshua Southern, Yam Eitan, **Guy Bar-Shalom**, Michael M. Bronstein, Haggai Maron, Fabrizio Frasca
(Submitted to ICML)

Topological Blindspots: Understanding and Extending Topological Deep Learning Through the Lens of Expressivity
Yam Eitan, Yoav Gelberg, **Guy Bar-Shalom**, Fabrizio Frasca, Michael M. Bronstein, Haggai Maron
Oral presentation
(ICLR 2025)

Selected Publications (* indicates equal contribution) (continued)

2024 A Flexible, Equivariant Framework for Subgraph GNNs via Graph Products and Graph Coarsening
Guy Bar-Shalom*, Yam Eitan*, Fabrizio Frasca, Haggai Maron
(NeurIPS 2024)
A shorter version appeared @ Symmetry and Geometry in Neural Representations workshop @ NeurIPS 2024 (oral) winning the best paper award

Subgraphormer: Unifying Subgraph GNNs and Graph Transformers via Graph Products
Guy Bar-Shalom, Beatrice Bevilacqua, Haggai Maron
(ICML 2024)

Weakly-supervised Representation Learning for Video Alignment and Analysis
Guy Bar-Shalom, George Leifman, Michael Elad
(WACV 2024)

2023 Window-Based Distribution Shift Detection for Deep Neural Networks
Guy Bar-Shalom, Yonatan Geifman, Ran El-Yaniv
(NeurIPS 2023)

2022 TransBoost: Improving the Best ImageNet Performance using Deep Transduction
Omer Belhasin*, **Guy Bar-Shalom***, Ran El-Yaniv
(NeurIPS 2022)

Awards

2025 Jacobs Qualcomm Excellence Scholarship

Faculty Excellence Scholarship, Technion

2024 best paper award @ Symmetry and Geometry in Neural Representations workshop @ NeurIPS 2024

VATAT Prize for Academic Excellence, Technion

2018 – present Teaching Assistant Excellence award (4 times).

Professional service

2024 Organizer of iGDL: Israeli Geometric Deep Learning workshop

2022-present Reviewer at NeurIPS, ICML, ICLR

Teaching experience

2018 – present Teaching Assistant at the Technion (department of Mathematics)
Course: *Partial Differential Equations and Fourier Series*
Average teaching score: 4.5/5.

2021 – present Teaching Assistant at the Technion (department of Computer Science)
Course: *Algorithms 1*
Average teaching score: 4.6/5.

2021 Teaching Assistant at the Technion (department of Computer Science)
Course: *Introduction to Computer Science*
Average teaching score: 4.4/5.

Other

2012	Played in the NCAA basketball league (USA), at Mercyhurst University (Erie, PA)
2012 – present	Semi-pro basketball player in Israel
2013-2016	Israel Defense Forces (IDF), Staff Sergeant